

Due Process on Hold: A Queueing Framework for Improving Access in SNAP

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The U.S. social safety net delivers essential services at mass scale, but access burdens persist, as congested contact or call centers serve as a primary mode of application completion and assistance. In *Holmes v. Knodell*, Missouri’s SNAP call centers were so congested that nearly half of all application denials were procedural—caused by applicants’ inability to complete required interviews, rather than underlying ineligibility. The judge ruled these system failures led to a violation of procedural due process. We propose a performance evaluation framework based on queueing models from operations research and management to assess and improve access in such systems. Operational access failures of call centers are distinct from prior automation failures in benefits provision. *Emergent arbitrariness* arises from interactions between system dynamics and access demand, rather than from an explicit algorithmic rule, making diagnosis and repair inherently system-level. We develop a queueing model that incorporates phenomena that distinguish social services from standard service domains, redials and abandonment, through which backlogs generate *endogenous congestion*. Using a fluid approximation, we derive steady-state performance metrics to analytically characterize the impacts of policy changes. We fit model parameters to call-center data disclosed in court documents. Our queueing model can support ex-ante evaluation and design of access systems, inform policy levers for improving access, and provide evidence about whether applicants are afforded a meaningful opportunity to be served at scale.

CCS Concepts: • **Applied computing** → **Operations research**; • **General and reference** → **Performance**; **Metrics**; • **Mathematics of computing** → **Queueing theory**.

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1 Introduction

Social safety net programs deliver essential services and benefits at a massive scale under major resource constraints. Operational challenges in service delivery can undermine efforts to achieve policy goals and program effectiveness. Recently, a district court case, *Holmes v. Knodell*, ruled that Missouri Department of Social Services’ (DSS) Supplemental Nutrition Assistance Program (SNAP) call center was so congested that enrollees were unable to call through to complete required interviews and applications, constituting a procedural due process violation. *Procedural denials* occur when an application is denied purely on procedural or administrative grounds, such as missing an interview, rather than the eligibility facts of the case. In Missouri, such procedural denials were as high as 50% of all denials. To the best of our knowledge, this is the first time that an *operational system (the call center) was found to violate procedural due process*. While the case ruling has concluded, the story is not complete. Missouri DSS has been ordered to improve call center operations under severe staffing constraints, reflecting a broader challenge across public-sector service delivery: how to provide meaningful access at scale with limited resources. How should Missouri DSS improve their call center design to provide *de facto* access to complete *de jure* application requirements?

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The operational failures of long wait times in automated systems are systemic both in *origin* and *future improvement*, differing from prior automation failures in benefits provision. We do not find available technical guidance that addresses the specific challenges of social service delivery. Queueing theory, which is a key subdiscipline of operations research (OR), is the natural tool for performance analysis. Call centers have been a particularly fruitful domain, as they feature the core ingredients of the field: large volumes of customers, random fluctuations in the times between arrivals and the durations of service, and strong potential for long wait times despite the risks of impatient customers. Indeed, federal guidelines for performance management of SNAP call centers refer to queueing theory and mention the famous Erlang-A formula, while also conceding that the *specific* nature of social services means that standard off-the-shelf models may not directly apply and the assumptions for this famous Erlang-A formula do not actually hold in practice [?].

Here, we develop a queueing model to fill the gap between standard call center models and special challenges in social services. Call centers for SNAP, and benefits more broadly, specifically suffer from *abandonment* where busy callers, some of whom may be paying by the minute [?], cannot stay on hold forever and may drop their call. But, different from more standard private-sector queueing models, callers applying to satisfy SNAP interview requirements who abandon are likely to call back, or *re-dial*, since SNAP and other social safety net programs serve urgent needs without private outside options. Moreover, the current *process design* of interviews often surface additional follow-up steps resulting in future calls. These behaviors, distinctive to social services, *compound operational deficiencies*, introducing stark feedback loops where prior backlogs result in abandonment that only worsens future congestion due to re-dials.

Our model can be helpful beyond the administration of SNAP alone. Other programs also rely on call and contact centers, just as major enterprises do, to serve as an interface with a massive client base. In practice, backlogs and queues in accessing government services comprise major crises in administration and service delivery. Unemployment insurance suffered major backlogs in disbursing benefits and call center waits during the Great Recession in 2008 and during the COVID-19 pandemic [? ? ?]. In 2025, backlogs grew for the Social Security Administration (SSA) amid chaotic changes in field operations at SSA offices and chronic understaffing [?]; agencies reported an average speed to answer that excluded the wait time for a callback (1-2 hours on average) while around 25% of calls were abandoned.

The focus of our work is in developing a performance evaluation and improvement framework, expanding upon classical queueing models from operations research, that can support efforts to improve access in social services. Our model, though reliant on some stochastic assumptions, allows us to estimate key performance metrics like wait times, abandonment (informative of procedural denials), throughput, and others, under changes in system parameters. The queueing model captures the *feedback loops* that arise when relying on call centers for required interviews and re-certifications that themselves generate follow-up contacts. Agencies could use our framework to understand trade-offs between different call center system design choices and resource expenditures. Although a theoretically simple fix is to just hire more staff until backlogs are cleared, in practice, inevitable resource, training and hiring constraints require exploring other system design changes. We introduce our queueing model, a fluid approximation, and discuss how to calibrate the model to data, which we do using court documents from *Holmes v. Knodell*. We introduce a simple dashboard to explore the impact of design changes on performance metrics. Our model admits analytical expressions for key performance metrics, which we analyze to obtain insights on how different design changes impact performance. Finally, we use our calibrated model to illustrate how an agency might combine staffing and system/policy design levers, like reducing average handling time or lengthening re-certification cycles, in order to achieve greater improvement with less staffing.

2 Background

Background on SNAP. The Supplemental Nutrition Assistance Program (SNAP), formerly the Food Stamp Program, is the largest means-tested social assistance program in the United States and provides monthly food-purchasing benefits to low-income households. SNAP is federally authorized and funded but administered by state and local agencies, which are responsible for determining eligibility and operating application and recertification processes within federal rules. In fiscal year 2024, “SNAP served an average of approximately 41.7 million individuals per month—about 12.3% of the U.S. population” [?]. Participation by state varies widely, from as high as 21.2% in New Mexico to 4.8% in Utah [?]. Studies of SNAP and other benefits programs find improvements not only in food security, but also health and labor outcomes, as well as stabilization against macroeconomic downturns [?]. Nonetheless, many who are eligible for SNAP do not receive benefits. Program eligibility is complex, including income determination and verification, deductions, asset limits, household definition, and work requirements — as is the application [?]. Such complexity introduces “administrative burden,” the requisite learning, psychological load, and compliance costs that citizens experience in interactions with government [?]. Many who are denied benefits because of operational frictions (missed interviews, application errors) are actually eligible [?].

The role of the interview, applications and recertification process: Call and contact centers remain one of the primary ways that potential enrollees interact with SNAP program administration [?].¹ Key administrative *checkpoints* [?] in the SNAP process include initial application, which requires an interview, and re-certification every 6 to 12 months, which also requires an interview. The SNAP recertification form measures changes of circumstances² which may change eligibility; any changes require submitted verification. During SNAP interviews, the caseworker may ask for additional documents to prove certain eligibility requirements or provide additional detail. Due to the complexity of SNAP eligibility, initial interviews also often serve as opportunities to inform applicants about complex SNAP-specific definitions. Some jurisdictions grant at least a period of at least 10 days to collect this additional information [?].

Missouri DSS has a waiver from Food and Nutrition Services (FNS), the federal agency administering SNAP, to implement interview procedures deviating from federal guidelines - namely, DSS does not need to schedule interviews, but must provide applicants with a notice (interview letter) instructing them to contact the Call Center within 5 days of submitting their application. If they do not, they receive a notice of missed interview informing that they need to complete the interview within 30 days of submitting their application.

More broadly, there are no uniform performance standards for operational performance for interviews. [?] comment on the legal context of *Holmes v. Knodell* and the challenge of rendering on-the-ground wait times legally cognizable. Existing enforceable laws did not anticipate the omnichannel service and access needs of today’s day and age. Although interviews are *required* by federal regulation to be completed within 30 days, federal regulation does not stipulate performance standards on service quality of *how* interview opportunities should be granted to enrollees. It remains unclear what *exact* metrics might lead to judicial findings of procedural due process violations³.

¹Services provided by call centers can provide everything from general information to official case services like updating and processing changes, conducting interviews, providing updates on processing, appointment scheduling and more. Staffing SNAP call centers is difficult because of SNAP policy complexity and the need to access live case information; call center staff touching the eligibility process must be merit system personnel [?]. As a result, there is large turn-over in call center staffing, chronic under-staffing, and it can take a long time to bring new staff up to speed [?] — Missouri DSS faced these challenges as well.

²for example, in lodging and housing costs, household composition, income, schooling status of income, food preparation household determination, income proof for individuals in food-preparing household, proof of benefits recipience, proof of changes in care/health expenses

³For example, the judge noted that in Missouri near 50% of denials were procedural. But other jurisdictions also have high rates of procedural denials that reflect persistent access barriers; for example, Los Angeles in 2018-2019 had a baseline rate of 30% procedural denials [?].

The grounds for the judicial decision: The judge concludes that the call center and administrative process (including low-staffed resource centers) to handle SNAP applications are overwhelmed. Plaintiffs contend that wait times are unacceptable, with some plaintiffs calling ten to twenty times and unable to call through, and legal services advocates waiting in the queue for hours before getting disconnected [?]. DSS is understaffed⁴. Thirty-two percent of all calls that made it into the interview-only queue were abandoned by the caller. Fifty percent of all applicants were denied for failure to complete the interview in one month during 2023, due only to a failure of the system to offer a reasonable opportunity to interview.

Per the order granting plaintiffs' motion [?], "Federal law requires Defendant to provide timely, accurate, and fair service to applicants for, and participants in, the SNAP program..." The changes, modifications, and adjustments made by DSS after increasing its reliance on its telephone system to process SNAP applications have proven unsuccessful in fulfilling Defendant's responsibility to provide timely, accurate, and fair service for applicants and participants in the program. The judge concludes [?] that the "reliance on an inadequate automated system and understaffed offices to provide interviews also violates Defendant's obligation under SNAP and Defendant's on-demand waiver," and therefore the corresponding automatic denial based on the automated system is a wrongful denial of benefits based on the arbitrariness of whether enrollees can call through. "Here, applicants are being denied benefits not based on the merits of their application but on the failure to obtain an interview. Further, the failure to interview is a direct result of Defendant's inability to provide an efficient and successful system that allows applicants to schedule and complete an interview within the required time frame."

Background on procedural due process: Much prior work in the algorithmic fairness community has focused on connecting algorithmic harms or performance legal theories of discrimination. *Holmes v. Knodell* is a procedural due process case, like other cases of algorithmic harms in benefits provision. *Procedural due process* requires that the state follow certain procedures when depriving someone of their property, such as giving notice, an opportunity to be heard (for example, in court), and a neutral decision-maker [?]. Courts have ruled, since the *Goldberg v. Kelly*, 1970, Supreme Court case, that social safety net and welfare programs represent significant property interests and hence are subject to procedural due process protections. Traditionally, such protections, drawing upon due process in courtroom procedures, might take the form of adversarial hearings, appeals, tribunals, or legal representation.

Scholars in administrative law have argued that such post-hoc appeals and hearings may not provide strong enough procedural protections for the massive scale of social safety net programs. [?] studies the Social Security Disability Insurance (SSDI) program and adjudication process and argues for the "management side of due process". Mashaw argues that burdensome appeals surface a small fraction of the agency's actual errors. Instead, fairness, accuracy, and timeliness are properties of *system performance*, not just rare appeals. Pro-active *continuous monitoring and performance management* can provide more systematic guarantees than individualized procedure that is overly reliant on ex-post correction. Fairness and accurate determinations are therefore emergent property of a system's performance as a whole, not necessarily individual procedural protections. For example, quality assurance at SSA comprises of random sampling for a second adjudication, as well as error analysis and monitoring. [? ?] connect these arguments about performance management and quality assurance programs in agencies with modern calls for AI auditing, algorithmic accountability, and performance management. [?] analyzes the role of administrative law in governing the emerging use of AI in benefits adjudication, with an example audit of recent state-deployed chatbots for assistance in unemployment insurance benefits.

⁴DSS has 200-300 staffers split in between informational (non-interview) calls and interview calls, while it estimates it needs 400 interview-only staffers to meet private call center "industry standards" of answering 80% of calls within 2 minutes

We share a similar high-level emphasis on *the role of performance evaluation in operationalizing due process in benefits provision*. However, we focus specifically on call center performance, which introduces distinct technical challenges.

Emergent systemic arbitrariness from automated systems. Automated systems have introduced great harms in crucial social safety net programs and benefits provision [? ? ?]. ?] overviews court dockets contesting algorithm-based determinations in disability, unemployment, and nutrition assistance programs and taxonomizes issues as flawed data, problematic system design choices, or inherent system limitations. *Holmes v. Knodell* introduces new challenges because the problematic service deficiencies, like long wait times and inability to reach a live representative, are systemic consequences of the automated system. It's not clear that there's a specific *system design choice* that violates procedural due process protections. We call this *emergent arbitrariness*.

We go into more detail on the judge's comparison to *Barry v. Lyon*⁵, when interpreting the SNAP Act's prohibition on wrongful denials, to highlight how the *emergent arbitrariness* differs from prior system design failures that resulted in arbitrary agent actions. In *Barry v. Lyon*, the issue is that Michigan Department of Health and Human Services (DHHS) introduced an additional automated match that added an additional eligibility requirement beyond those federally mandated, violating the SNAP Act. But then the fix is simple — Michigan DHHS should remove the automated match. In contrast, for SNAP call centers, the *origins of* and *future improvements* for the *emergent arbitrariness* are more complex than a single system design flaw. The judge can mandate that Missouri DSS must modify its administration process to ensure the opportunity to interview, but it remains unclear *how* they should do this. A first question is - what exactly is the automated system at fault here? On the one hand, one might object to the *automatic denial* of applications in 30 days, even from those who try to call in to interview, and keep applications alive for those who try to call in (stay the denial). But without also improving system throughput, staying application denial without improving wait times only delays the ultimate approval of crucial benefits to enrollees. This leaves the locus of automation concerns to the automated aspects of the system that grants interviews to callers, the call center's routing and design. But we argue that the *systemic long wait time outcomes* of the call center emerge from complex interactions between demand (inflow of individuals applying for or recertifying SNAP), system design (automated procedures that dictate how incoming calls are routed), and caller behavior (re-dialing when unable to connect for an interview or abandoning calls). We distinguish *emergent arbitrariness* from failures in which algorithmic rules directly violate *de jure* procedural due process (e.g., defective notice or fully automated determinations).

3 Related Work

Informal and formal performance evaluation of government benefits systems: In formal quality control processes for SNAP, FNS collects random samples of household case files and analyzes eligibility determinations and overpayments/underpayment errors [?]. However, these formal performance evaluations focus on payment errors, rather than other system performance measures including administrative burdens, access, or other measures of otherwise-eligible households that nonetheless do not complete applications or receive benefits. Regarding call center performance specifically, there is wide heterogeneity in evaluation structure and specific metrics used across different government agencies and programs. (See ?] for an overview of practices and metrics used for unemployment insurance⁶). However,

⁵In *Barry v. Lyon*, the court ruled that Michigan Department of Health and Human Services must stop automatic denial of SNAP applications based on an automated match with a list of outstanding felony warrants (rather than those actively fleeing/actively being sought by law enforcement), which included erroneous matches such as aliases given by others or cases of identity theft.

⁶The Department of Labor contracted a consulting firm to survey different state leaders about their operational practices in call centers for unemployment insurance. They found wide variation in exact operational practices, but common use of strategies like cross-training and pooling queues across multiple call centers, while finding that many states could use more forecasting and queue management tools [?].

the IRS has long provided assistance to tens of millions of Americans by phone, and has long designed its own call center quality and customer experience evaluations [?]. In the absence of formal performance assessments and real-time reporting, outsiders may also run informal audits to report performance metrics. [?], in a "mystery shopper" approach, recruited research assistants to call various state unemployment call centers during the COVID-19 pandemic to assess how long it took to get a live representative on the line, to quantify the backlog and access issues.⁷ Sen. Elizabeth's Warren's office made hourly calls for a week, finding average wait times of around 100 minutes and that only 50% of calls connected to a live representative without being dropped by SSA [?].

Administrative burdens and SNAP: Recent works study changes in interview operations that introduce or improve administrative burdens, demonstrating beneficial impacts of interview flexibility on approvals and long-term participation [? ?] or barriers from less time to reschedule [?].⁸

Other operational analysis of public-sector queueing systems: Queueing analysis is a mature discipline, and prior studies have also focused on public sector systems, albeit not the SNAP access system here. Some prior studies focus on operations in the judicial case management system. [?] study the queueing system in immigration courts and analyze fairness/efficiency properties of the queueing discipline, as well as strategic considerations specific to when waiting provides utility [?]. [?] study delays in judicial case management. Judicial case management differs from our eligibility/administrative burden setting: in court docket management, cases stay "live" for extended periods of time, while for social services, the interview requirement must be completed within 30 days. [?] study information design in a general model to target high-need individuals without outside options to manage a congested queue for social services.

Other technical work and benefits provision: [?] provides an overview of civic tech and digital services in the federal government, including key benefits tech infrastructure. [?] audits benefits *eligibility calculators* for potential errors. [?] explores AI and trust in LLM chatbots supporting SNAP applicants. [?] audits ad allocation budgets for SNAP outreach and in a survey, finds broad support for equity considerations.

4 Queueing Model

While the conventional approach to queueing models of call centers does recognize that customers may leave the queue before reaching a service agent, they typically assume that abandoning customers are lost forever [?]. Similarly, customers who complete service are typically thought of as departing the system once and for all. Neither of these assumptions are well-aligned with the SNAP context: because this is an essential service, customers who abandon the queue are very likely to call back later, and, because the benefits require regular re-certification, successfully served customers are due to eventually return.

With those dynamics in mind, we build from prior work on *re-dials* and *re-connects* in [?] to capture four different possible paths, or "orbits," in which callers may return to the SNAP call center again after abandoning or completing service, with two "orbits" from abandonment and two "orbits" from completed service.⁹ For abandonment, the case data and documents show that not only are abandoning callers quite likely to call back, there are different cadences to when those re-dials occur. Some abandoning callers call back on the order of hours later, whereas others call back a number of days later. We deem these short and long re-dial orbits, respectively. These re-dial orbits highlight the *negative*

⁷Earlier iterations of the IRS' call center evaluation included "mystery-shopper" like programs, with the downside that queries are artificial rather than actual customer concerns [?].

⁸[?] find that later recertification interviews, with less time for rescheduling, decrease recertification success, including for truly eligible clients. [?] study a field experiment of applicant-initiated *flexible* on-demand interviews in Los Angeles, increasing approvals by 6% and long-term participation by 2%. [?] run a field experiment comparing text-message reminders of flexible interviews vs. mailed reminders in Boulder County, Colorado, finding earlier and 10% more likely interview completion.

⁹In practice, lost customers includes those who abandoned, disconnected, or were blocked, which we collapse into one phenomenon in the model.

feedback loops in congested social services. Then, for callers who are able to get connected to an agent and complete their call, we view two possible paths: either the call was successful, and the caller becomes enrolled in benefits, or the call was not successful, and the caller will eventually need to call back. For the latter, we direct unsuccessful service completions into the long re-dial orbit, and, for the former, we direct to an even longer returning orbit, on the order of months, to represent the eventual need to re-certify eligibility for benefits. To that end, we model two possible true exits from the system. First, on the enrolled orbit, we allow some probability that the recipients will not pursue their next re-certification; such attrition can arise from exogenous events that remove households from eligibility. Second, we suppose that there is some fraction of abandoning callers who truly do not ever call back, do not enter any orbit, and are thus lost from the system – these are the model’s procedural denials.

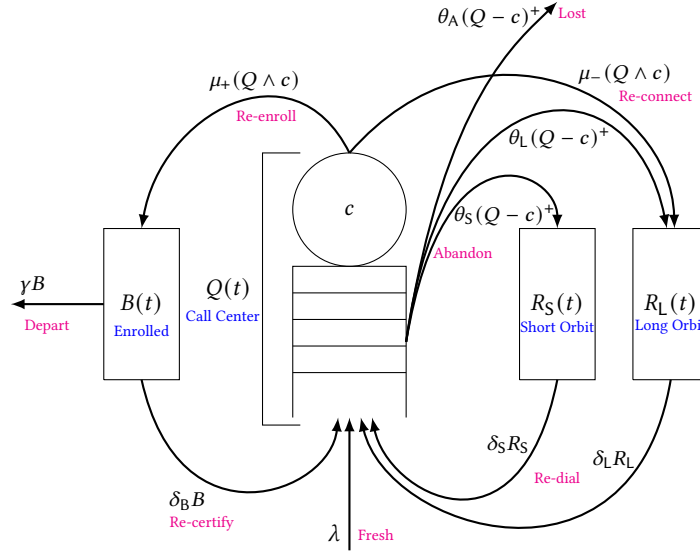


Fig. 1. Process flow diagram of a queueing theoretic model of the SNAP call center.

Definition of the SNAP call center stochastic model. In a manner closely following [?] but with further details added for the sake of the problem context, let us model the call center as a Markovian service network. First, let $Q(t)$ be the number of callers either actively speaking to one of the $c \in \mathbb{Z}_+$ agents or waiting for the next available agent at time $t \geq 0$. Let $\lambda > 0$ be the rate of “fresh” or exogenous arrivals to the call center for those newly seeking to enroll in benefits. Suppose that each agent completes calls at rate $\mu = \mu_+ + \mu_-$ for $\mu_+, \mu_- > 0$, where each completed call has a probability μ_+/μ of being successfully completed, meaning that the caller will not need to call back in order to complete enrollment and receive benefits, and probability μ_-/μ of being unsuccessful, meaning that the caller will need to call back. Let $B(t)$ track the number of successfully enrolled recipients at time t . Successfully enrolled recipients will either eventually “qualify out” of the system or otherwise need to re-certify. At rate $\delta_B B(t)$, $B(t)$ will decrease by one with $Q(t)$ simultaneously increasing by one, and at rate $\gamma B(t)$, $B(t)$ will simply be decremented.

Now, we will suppose that callers waiting for the next available agent each have independently and exponentially patience time with rate $\theta = \theta_A + \theta_S + \theta_L > 0$ for $\theta_A, \theta_S, \theta_L \geq 0$. With probability θ_A/θ , the caller will abandon the queue and never call back to re-attempt enrollment. Then, with probability θ_S/θ , the caller will hang up but eventually “re-dial”

or attempt to call back after a (likely) short amount of time. Let $R_S(t)$ be the number of callers at time t who have left the queue but will soon re-dial, where each re-dial time is independent and exponentially distributed with rate δ_S . Finally, with probability θ_L/θ , a caller will hang up and eventually call back but after a long time like the unsuccessful callers; hence, we count both types of callers among $R_L(t)$. On aggregate, at rate $\delta_S R_S(t)$, $R_S(t)$ will decrease by one with $Q(t)$ increasing by one, and, likewise, at rate $\delta_L R_L(t)$, one caller from $R_L(t)$ will transition back to $Q(t)$.

In this way, the full model (Q, B, R_S, R_L) is a multi-station queueing network where the Q station is a $\cdot/M/c$ queue and each of B , R_S , and R_L are $\cdot/M/\infty$ queues. We visualize this queueing network in Figure ?? . Formally, the stochastic process (Q, B, R_S, R_L) can be constructed through Poisson processes:

$$\begin{aligned} Q(t) = Q(0) &+ \Pi_\lambda(t) + \Pi_B\left(\int_0^t \delta_B B(s) ds\right) + \Pi_S\left(\int_0^t \delta_S R_S(s) ds\right) + \Pi_L\left(\int_0^t \delta_L R_L(s) ds\right) \\ &- \Pi_{Q:A}\left(\int_0^t \theta_A (Q(s) - c)^+ ds\right) - \Pi_{Q:S}\left(\int_0^t \theta_S (Q(s) - c)^+ ds\right) - \Pi_{Q:L}\left(\int_0^t \theta_L (Q(s) - c)^+ ds\right) \\ &- \Pi_{Q:+}\left(\int_0^t \mu_+ (Q(s) \wedge c) ds\right) - \Pi_{Q:-}\left(\int_0^t \mu_- (Q(s) \wedge c) ds\right), \end{aligned} \quad (1)$$

$$B(t) = B(0) + \Pi_{Q:+}\left(\int_0^t \mu_+ (Q(s) \wedge c) ds\right) - \Pi_B\left(\int_0^t \delta_B B(s) ds\right), \quad (2)$$

$$R_S(t) = R_S(0) + \Pi_{Q:S}\left(\int_0^t \theta_S (Q(s) - c)^+ ds\right) - \Pi_S\left(\int_0^t \delta_S R_S(s) ds\right), \quad (3)$$

$$R_L(t) = R_L(0) + \Pi_{Q:-}\left(\int_0^t \mu_- (Q(s) \wedge c) ds\right) + \Pi_{Q:L}\left(\int_0^t \theta_L (Q(s) - c)^+ ds\right) - \Pi_L\left(\int_0^t \delta_L R_L(s) ds\right). \quad (4)$$

Here, as in the literature [e.g., ??], we use $\Pi_i(\cdot)$ for $i \in \mathcal{I} = \{\lambda, B, S, L, Q : A, Q : S, Q : L, Q : +, Q : -\}$ are mutually independent unit-rate Poisson processes. We assume that $Q(0)$, $B(0)$, $R_S(0)$, and $R_L(0)$ are known initial conditions.

Model interpretability via Poisson thinning and superposition. Because the queueing network model is composed from a collection of Poisson processes, we can leverage well-known probabilistic properties to further interpret its transitions. For example, consider the dynamics of the number of presently enrolled recipients. Intuitively, the next decrease to $B(t)$ will be either a departure via attrition from the benefits system or a return to the call center to re-certify. Because of the memoryless property of the underlying exponential random variables between events in Poisson processes, we can view this as a “race” between two exponentially distributed “clocks” – one for attrition and one for re-certification. Moreover, because of the thinning property of the Poisson itself, we can granularize this same interpretation to the level of each currently enrolled individual: for each current benefit recipient, there is a $\text{Exp}(\gamma)$ clock for the time to potential attrition and a $\text{Exp}(\delta_B)$ clock for the time to potential re-certification, and whichever clock rings first will be what happens next. Hence, by these same properties, there is a $\gamma/(\gamma + \delta_B)$ that the recipient departs via attrition and a $\delta_B/(\gamma + \delta_B)$ that they instead call to re-certify. Similar interpretations abound throughout the model: an abandoning caller enters the short orbit with probability $\theta_S/(\theta_A + \theta_S + \theta_L)$, a concluding caller enters the long orbit with probability $\mu_-/(\mu_+ + \mu_-)$, and each of the $(Q(t) \wedge c)$ ongoing calls are equally likely to be the next to conclude.

4.1 Fluid model

As a simpler companion to the stochastic queueing model, let us also define the deterministic *fluid model* (q, b, r_S, r_L) through the following system of ordinary differential equations (ODEs):

$$\dot{q}(t) = \lambda + \delta_B b(t) + \delta_S r_S(t) + \delta_L r_L(t) - \theta(q(t) - c)^+ - \mu(q(t) \wedge c), \quad \dot{b}(t) = \mu_+(q(t) \wedge c) - (\delta_B + \gamma)b(t), \quad (5)$$

$$\dot{r}_L(t) = \mu_-(q(t) \wedge c) + \theta_L(q(t) - c)^+ - \delta_L r_L(t), \quad \dot{r}_S(t) = \theta_S(q(t) - c)^+ - \delta_S r_S(t). \quad (6)$$

As one may recognize by comparison to Figure ??, the dynamics of these ODEs align with the overall flow of the queueing model. However, the fluid model is deterministic, unlike the stochastic queueing model, suggesting that the fluid model could be a good first-order approximation for the queue. There is a clean intuition for that approximation: Imagine the call center queue as a physical system, where callers actually arrive to process through the network as visualized in Figure ?. Now, imagine those customers as arriving faster and faster, but also proportionally shrinking smaller and smaller, so that they move through the various services and orbits faster and faster as well. As this shrinking and speeding goes to the extreme, the queueing network would look like a continuous flow of callers in and out of the stations of the queue, where the random times between arrivals, abandonments, completed calls, re-dials, re-connects, and re-enrolls would be replaced by fluid at the same rates. Moreover, following intuition granted by the law of large numbers, the stochastic fluctuations brought by those random arrival, abandonment, and service times should eventually be dominated by the sheer scale of the growing and accelerating system.

This intuition can be formalized: the fluid model indeed arises out of a functional strong law of large numbers (FSLLN) limit of the queueing model, and we now formalize this connection in Proposition ??.

PROPOSITION 4.1. *For $n \geq 0$, let $(Q^{(n)}, B^{(n)}, R_S^{(n)}, R_L^{(n)})$ be the queueing network model of (??) through (??) alternately defined with external arrival rate λn and number of servers cn . Then, given that*

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n} Q^{(n)}(0), \frac{1}{n} B^{(n)}(0), \frac{1}{n} R_S^{(n)}(0), \frac{1}{n} R_L^{(n)}(0) \right) \longrightarrow (q^0, b^0, r_S^0, r_L^0), \quad (7)$$

the fluid limit of the scaled stochastic process converges to the deterministic system,

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n} Q^{(n)}(t), \frac{1}{n} B^{(n)}(t), \frac{1}{n} R_S^{(n)}(t), \frac{1}{n} R_L^{(n)}(t) \right) \xrightarrow{\text{a.s.}} (q(t), b(t), r_S(t), r_L(t)), \quad (8)$$

uniformly on compact sets, where $(q(t), b(t), r_S(t), r_L(t))$ is the unique solution to the system of equations in (??) through (??) with initial condition $(q(0), b(0), r_S(0), r_L(0)) = (q^0, b^0, r_S^0, r_L^0)$.

PROOF. This fluid limit can be readily obtained from prior functional strong law results from the literature; for instance, the proof of this limit follows immediately from theorem 2.2 of [?], where section 5 therein provides a relevant specific application of that general FSLLN result. $\square$

From this combination of intuition and rigorous connection, fluid models are very commonly used to study queueing models in a simplified manner that grants tractability but still captures the heart of the modeling context.¹⁰ These benefits are particularly evident when studying the steady state of the system, in which the system of ODEs simplify to a system of linear equations. By setting the equations in (??) and (??) each to 0 and solving for an equilibrium solution, which we denote $(\bar{q}, \bar{b}, \bar{r}_S, \bar{r}_L)$, this fluid model admits a first-order approximation for the system's steady-state means:

$$\bar{q} = \left(c \wedge \left(1 + \frac{\delta_B}{\gamma} \right) \frac{\lambda}{\mu_+} \right) + \frac{1}{\theta_A} \left(\lambda - \frac{\gamma c \mu_+}{\gamma + \delta_B} \right)^+, \quad \bar{b} = \left(\frac{\lambda}{\gamma} \wedge \frac{c \mu_+}{\gamma + \delta_B} \right), \quad (9)$$

$$\bar{r}_L = \left(\frac{c \mu_-}{\delta_L} \wedge \left(1 + \frac{\delta_B}{\gamma} \right) \frac{\lambda \mu_-}{\delta_L \mu_+} \right) + \frac{\theta_L}{\delta_L \theta_A} \left(\lambda - \frac{\gamma c \mu_+}{\gamma + \delta_B} \right)^+, \quad \bar{r}_S = \frac{\theta_S}{\delta_S \theta_A} \left(\lambda - \frac{\gamma c \mu_+}{\gamma + \delta_B} \right)^+. \quad (10)$$

Distinguishing arrival rates in the fluid model. By comparison to the *fresh* arrival rate, λ , which only counts callers that are newly seeking benefits, the overall rate of arrivals to the system will also include re-dialing callers and

¹⁰Critically, let us note that the solutions to the fluid model equations need not exactly match the underlying means of the stochastic model, especially for queues with abandonment [e.g., ?]. Nevertheless, for large systems with many arrivals and many servers (as in our motivating context), the stochastic model is typically close enough to the fluid approximation for its first-order insights to be both valid and valuable, as implied by Proposition ??.

current benefit recipients seeking to re-enroll. Because the total arrival rate is both an important quantity in its own right and more readily observed in data, let us formalize it in notation. Specifically, in the fluid model, let $\hat{\lambda}(t) = \lambda + \delta_B b(t) + \delta_S r_S(t) + \delta_L r_L(t)$ denote the total arrival rate to the system at time t , and, without the t argument, let $\hat{\lambda} = \lambda + \delta_B \bar{b} + \delta_S \bar{r}_S + \delta_L \bar{r}_L$ be the total arrival rate for the fluid model in steady-state.

Furthermore, in studying the caller experience, it will also be of interest to focus on the volume of arrivals that includes the re-dialing and re-enrolling orbits but excludes the customers that renege from the queue before connecting to an agent. To be able to compute values such as the average waiting time among callers that do not abandon, let us define $\tilde{\lambda}(t) = \hat{\lambda}(t) - \theta(q(t) - c)^+$ as the total arrival rate at time t less abandonments, and we again let the argument-less $\tilde{\lambda}$ denote this quantity in steady-state.

Simplifications in the overloaded regime. When the system is overloaded, i.e. $q(t) \geq c$, then we can simplify the branching logic so that $(q(t) - c)^+ = q(t) - c$ and $(q(t) \wedge c) = c$. Moreover, in steady-state, the overloaded regime can be easily characterized, and it yields simple expressions for the four model quantities.

COROLLARY 4.2. *The overloaded regime, $\bar{q} \geq c$, occurs as the equilibrium fluid model solution if and only if $\lambda \geq \gamma c \mu_+ / (\gamma + \delta_B)$, which equivalently occurs if and only if $\hat{\lambda} \geq c \mu$. In this case, the steady-state solutions simplify to*

$$\bar{q} = \frac{1}{\theta_A} \left(\lambda - \frac{\gamma c \mu_+}{\gamma + \delta_B} \right) + c, \quad \bar{b} = \frac{c \mu_+}{\gamma + \delta_B}, \quad \bar{r}_S = \frac{\theta_S}{\theta_A \delta_S} \left(\lambda - \frac{\gamma c \mu_+}{\gamma + \delta_B} \right), \quad \bar{r}_L = \frac{c \mu_-}{\delta_L} + \frac{\theta_L}{\delta_L \theta_A} \left(\lambda - \frac{\gamma c \mu_+}{\gamma + \delta_B} \right). \quad (11)$$

In the remainder of the paper, we will largely focus on the steady-state fluid model and assume it to be in the overloaded regime. This parsimonious approximation offers the simplest first step to derive insights from the *Holmes v. Knodell* data and assess the impacts of possible changes to the SNAP call center system. The overloaded assumption is well-justified, here and more broadly, since system-level stresses motivate us to study the problem in the first place.

5 Data and Model Fitting

In this section, we illustrate how to connect our model parameters to call center data in general, and we discuss how we do so for Missouri DSS' call centers via the aggregated data released in court documents.

Simple parameter matching from observable metrics: The eleven parameters used by the queueing model may not actually all be readily identifiable in practice. For instance, the model uses λ , the rate of *fresh* arrivals only, as a parameter, whereas $\hat{\lambda}$, the total rate of all arrivals, is likely what is recorded operationally either by hand or by typical call center management software. Rather, disentangling λ from $\hat{\lambda}$ is actually a possible benefit of the model.

Let us consider what quantities should be observable in practice and connect them to the (overloaded) steady-state fluid approximation.¹¹ First, as we have said above, we would expect that total arrival rate, $\hat{\lambda}$, to be known to the management of the call center. Similarly, we expect the service rate, μ , or the more recognizable $1/\mu$ the average call duration is readily observable.¹² We will also take the staffing level, c , as known. We furthermore expect that the time from benefit enrollment to re-certification is known, which gives δ_B via $1/\delta_B$ as the time until re-certification begins, and we likewise expect the benefits retention to be known, which gives γ via $\delta_B/(\gamma + \delta_B)$ as the retention rate.

We expect $\bar{q}$ and $\bar{b}$, the average number of present callers (including both waiting and in service), $\bar{q}$, and number of recipients presently enrolled in a benefits program like SNAP, are taken as known.

¹¹If fine-grained data on arrival times and call durations is available, more rigorous techniques like maximum likelihood estimation can deliver high fidelity parameter estimation. But since the fluid model is itself already a deterministic approximation to the queueing model's stochastic approximation of reality, these back-of-the-envelope calculations can be used to guide calibration of the fluid model from aggregated data.

¹²This service time *excludes* waiting and its calculation *excludes* any callers who abandon before being connected to an agent.

Assuming that the system is in the overloaded regime, we can take the above assumptions of observable quantities within the equilibrium fluid model and the definitions of $\hat{\lambda}$, θ , and μ to provide a system of equations for the remaining model quantities. We do so now, with unknown terms underlined for emphasis in each equation:

$$\begin{aligned}
 0 &= \underline{\lambda} + \delta_B \bar{b} + \underline{\delta_S \bar{r}_S} + \underline{\delta_L \bar{r}_L} - \theta(\bar{q} - c) - c\mu, & 0 &= c\underline{\mu_+} - (\gamma + \delta_B)\bar{b}, \\
 0 &= \underline{\theta_S}(\bar{q} - c) - \underline{\delta_S \bar{r}_S}, & 0 &= c\underline{\mu_-} + \underline{\theta_L}(\bar{q} - c) - \underline{\delta_L \bar{r}_L}, \\
 \hat{\lambda} &= \underline{\lambda} + \delta_B \bar{b} + \underline{\delta_S \bar{r}_S} + \underline{\delta_L \bar{r}_L}, & \theta &= \underline{\theta_A} + \underline{\theta_L} + \underline{\theta_S}, & \mu &= \underline{\mu_-} + \underline{\mu_+}.
 \end{aligned} \tag{12}$$

As written, this system has seven equations and ten unknowns. Hence, we need more information to solve for the missing values, albeit perhaps not as much as it might seem. First, consider δ_S , δ_L , $\bar{r}_S$, and $\bar{r}_L$. Several approaches can quickly reduce these four to just two missing values.¹³ Taking such reduction for granted, we are left with seven linear equations and eight unknowns. We turn to the abandonment dynamics to pin down this final degree of freedom. In the best case, in practice, there may be some domain knowledge that delivers these quantities, e.g., knowing the fraction of re-dialing callers that do so sooner rather than latter, which is modeled as $\theta_S/(\theta_S + \delta_L)$. At the other extreme, if little is known about the abandonment or re-dialing dynamics, one could accordingly reduce the model complexity by setting $\theta_S = 0$, which would acknowledge the ambiguity and simplify the model to have only one orbit (i.e., $\bar{r}_S = 0$ if $\theta_S = 0$). Finally, if neither of those two extremes match the context, one could simply sweep over possible values of, say, θ_A , and compare the resulting solution to some additional data or other outside information to determine the most reasonable option; we take this sweeping approach in our own analysis.

Parametrizing the Queueing Model from Aggregated Call Center Data: We now discuss how we fit parameters specifically for the aggregated call center and arrival data for court documents from *Holmes v. Knodell*, see ?? for more detail. We downloaded the court documents from PACER, which include extensive reports (typically monthly, weekly or daily) on arrival rates, call center staffing, and various other performance metrics like average handling time and average wait time.

We parametrize everything in terms of the number of minutes a call center is open in a day, $MMin = 540min = 9$ hours; the call centers are open on weekdays. In our model, some *operational* parameters such as the total arrival rate $\hat{\lambda}$, staffing levels c , and average handling time (AHT) are directly observed in the call-center data. To reduce parameter dimensionality, we fix a proportion $p_+ \in [0, 1]$ of completed calls that result in completed re/-enrollments, hence $\mu_+ = p_+ \left(\frac{MMin}{AHT} \right)$. We set $\delta_B, \delta_S, \delta_L, p_+$ by using a combination of program design, domain knowledge, and time series analysis. From SNAP eligibility guidelines, we fix the recertification period at 6 months (calendar days) ≈ 128.5 model (call center) days (since it's only open weekdays). This fixes the parameter $\delta_B = 1/128.5$. We set $\delta_S = 3$, corresponding to a typical callback time of 3 hours, based on domain context. The remaining behavioral parameters governing abandonment and re-dial behavior, $(\lambda, \theta_A, \theta_S, \theta_L)$, namely the fresh arrival rate λ and rates of total abandonment θ_A , short/long orbit abandonment θ_S, θ_L , are not directly observed in our aggregated call center data.¹⁴

Derived metrics and an operational dashboard: The steady-state approximation allows us to derive *model-based estimates* of performance metrics. The model results best inform *relative comparisons* as to how changes in call center design *increase or decrease key metrics*. We introduce the resulting *model-based performance metrics*, which use the

¹³For instance, in our data set, we use an auto-regressive time series analysis on the daily call volumes to suggest values for δ_S and δ_L , which then simply leaves $\bar{r}_S$ and $\bar{r}_L$ as unknown. On the other hand, if such data analysis is not possible, though, one can simply abstract away from viewing the δ individual rate and $\bar{r}$ average number in orbit separately to a single metric for the *volume* of returning callers per unit of time, say $\bar{v}_S = \delta_S \bar{r}_S$ and $\bar{v}_L = \delta_L \bar{r}_L$. Even though this abstraction would mean that we would lose some amount of visibility into the model overall, it would not compromise our access to evaluating key performance metrics: the degree of procedural denials and the volume of endogenous congestion can still be evaluated in this setting.

¹⁴From data, we only have measures of aggregated abandonment percentages (typically daily or weekly).

CC	c (staffing)	AHT (min)	Wait (min)	ASA (min)	observed $\hat{\lambda}$	best-fit λ	Best MAD ($\bar{w}$, $\tilde{w}$)
1	32.5	20.8	4.4	4.1	228.0	145.4	(1.2, 1.9)
2	52.0	21.9	38.8	117.8	2931.5	623.3	(2.9, 27.0)
3	72.0	44.2	45.5	87.6	2396.5	700.1	(11.6, 20.8)
4	51.0	44.8	1.8	1.8	712.0	164.8	(1.1, 1.6)

Table 1. Descriptive statistics on observed metrics for call centers (CC), and best-fit lambda values and MAD for each call center. Staffing and arriving calls are daily metrics.

approximated steady state values and include: average wait time ($\bar{w} = (\bar{q} - c)^+ / \hat{\lambda}$, which includes abandonments), average speed to answer ($\tilde{w} = (\bar{q} - c)^+ / \tilde{\lambda}$, which does not include abandonments), abandonment fraction ($1 - \frac{\tilde{\lambda}}{\hat{\lambda}}$), total lost abandonment ($\theta_A(\bar{q} - c)^+$), and endogenous congestion from re-dials ($\delta_S \bar{r}_S + \delta_L \bar{r}_L$).

We can interpret the metric of lost calls, $\theta_A(\bar{q} - c)^+$, those who completely leave the system, as a metric related to procedural denials. The fluid model is not sharp enough to estimate whether a single caller would get through in 30 days, so the total abandonment is a lower bound on procedural denials. Other relevant metrics include utilization $U = \frac{\hat{\lambda}}{c(\mu_+ + \mu_-)}$, describing the fraction of time service providers are busy, and the implied utilization $IU = \frac{\hat{\lambda}}{c(\mu_+ + \mu_-)}$.

We develop an analytics dashboard where stakeholders could perturb system parameters to examine impacts on these performance metrics, see ?? in the Appendix for a screenshot. The (anonymized version of) the dashboard is publicly available at <https://call-center-simulator-production.up.railway.app>.

Robustly fitting λ over a range of plausible $\theta_A, \theta_S, \theta_L$: Since the finer-grained behavioral parameters $\theta_A, \theta_S, \theta_L$ are uninformed from data, we fit a single fresh arrival rate parameter λ that recovers wait times well over a *potential range of plausible values of $\theta_A, \theta_S, \theta_L$* . We assess how the derived wait times $\bar{w}, \tilde{w}$ compare to *observed* wait times (including abandonment) and speed to answer (excluding abandonment). There are four different call centers in Exhibit 87. We group them by call volume and average handling time, two in a 20 minute AHT category and two in a 40 minute AHT category. We calculate the median average wait time and median average speed to answer. We sweep over a range of *reasonable $\theta_A, \theta_S, \theta_L$* ; we fix the sum of rates $\theta_A + \theta_S + \theta_L = 10$ or 54 (corresponding to 54 or 10 minute abandonment, respectively) and vary the relative weight of $\theta_A, \theta_S, \theta_L$. Finally, we robustly fit λ by minimizing the mean average deviation (averaging over θ parameter vector values) of median wait times.

Observed metrics and fitted parameters: This table summarizes the key metrics for the four call centers, separating the empirically observed data from the fitted model parameters. The columns from “c (staffing)” through “observed $\hat{\lambda}$ ” represent the direct operational measurements. All time metrics are shown in decimal minutes. Specifically, “Average Wait Time (min)” and “ASA (min)” correspond to the observed values for $\bar{w}$ and $\tilde{w}$, respectively, while “observed $\hat{\lambda}$ ” is the total arrival rate including re-dials. In contrast, the “best-fit λ ” is a fitted parameter, representing the optimized robust rate of fresh arrivals derived from the model. The final column, “Best MAD ($\bar{w}$, $\tilde{w}$)”, quantifies the result of this fit by showing the final Mean Absolute Deviation between the model’s predicted wait times and the observed “Average Wait Time” and “ASA Time”, respectively.

6 Evaluating system design changes

Analytical results for operational changes in the general fluid model. We begin by leveraging closed-form solutions for steady-state expressions for the performance metrics to analyze *how* different potential operational changes affect system performance in the overloaded regime. Specifically, we now analyze six key steady-state performance metrics: the procedural denial rate ($PD = \theta_A(\bar{q} - c)$), the mean number of waiting callers ($\bar{q} - c$), the mean waiting time ($\bar{w} = (\bar{q} - c) / \hat{\lambda}$),

the average speed to answer ($\tilde{w} = (\bar{q} - c)/\tilde{\lambda}$), the endogenous congestion from re-dials ($EC_R = \delta_S \bar{r}_S + \delta_L \bar{r}_L$), and the endogenous congestion from re-certification ($EC_B = \delta_B \bar{b}$). For each quantity, we evaluate its dependence on four model parameters that could reasonably be subject to managerial control or operational design: the staffing level (c), the frequency of re-certification (δ_B), the individual service rate (μ), and the fraction of successful calls (which we denote $p_+ = \mu_+/\mu$, with μ assumed to be held fixed).¹⁵

Intuitively, while c may be expensive to increase, it may be relatively easy to re-design the eligibility interview process to be either faster (increase μ) or more likely to result in certification (increase p_+), and, likewise, structural policy changes could increase the re-certification period (decrease δ_B). Here, we leverage the fluid model to offer insight of how these prospective changes may relate to one another. In the interest of space, the exact expressions of the partial derivatives of each performance metric with respect to each operational parameter are left for the appendix (see Propositions ?? through ??), and here we simply highlight their takeaways in a series of corollary results. We begin with the following review of the directional dependence of each performance metric on each operational parameter.

First, we find that any of the prospective changes will yield improvements in the procedural denial rate, the waiting time, the average speed to answer, and the number waiting.

COROLLARY 6.1. *In the overloaded regime, the procedural denial rate, mean waiting time, average speed to answer, and mean number waiting are each decreasing with c , μ , and p_+ and increasing with δ_B .*

Then, we similarly find that the volume of arriving callers seeking to re-certify will increase with higher staffing, faster service, better conversion rates, or shorter re-certification periods.

COROLLARY 6.2. *In the overloaded regime, the endogenous congestion from re-certifications is increasing with each of the operational parameters.*

On the other orbit, the endogenous congestion from re-dials decreases with better conversion or shorter re-certification periods, but, interestingly, it could either increase or decrease with higher staffing or faster service.

COROLLARY 6.3. *In the overloaded regime, the endogenous congestion from re-dials is increasing with δ_B and decreasing with p_+ . However, the endogenous congestion from re-dials increases with c and μ if $\theta_A(1 - p_+)(\gamma + \delta_B) > p_+\gamma c(\theta_S + \theta_L)$, and otherwise it is non-increasing with c and μ .*

Now, to give a sense of the relative scale of change, and not simply the direction of change, let us consider the elasticity of the performance metrics with respect to the operational parameters. For a metric y and parameter x , the (local) elasticity is defined as $(x/y)\partial y/\partial x$. Here, we find that, for each of the performance metrics, several parameters actually admit equivalent elasticities.

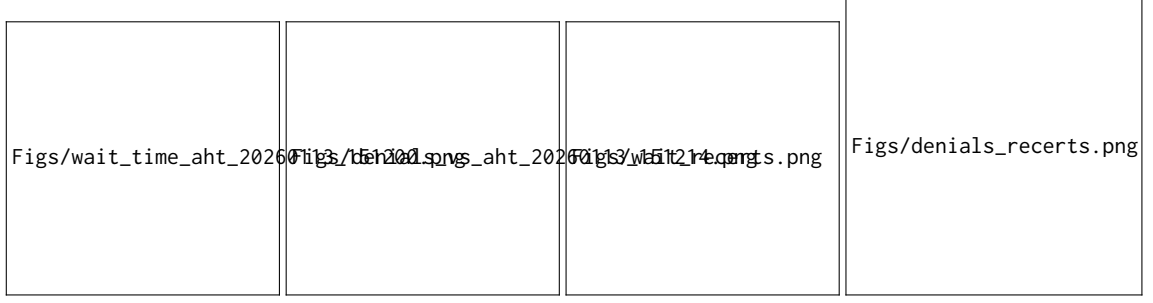
First, we find that changes to staffing level, service rate, and probability of success all yield the same respective elasticity for the procedural denial rate, number waiting, and endogenous congestion from re-certifications.

COROLLARY 6.4. *For each of PD, $\bar{q} - c$, and EC_B , the elasticity is the same across c , μ , and p_+ in the overloaded regime.*

Then, for the waiting time, speed to answer, and endogenous congestion from re-dials, changes to the staffing level and to the service rate again yield equivalent elasticities, but they need not be the same as that of the success probability.

COROLLARY 6.5. *For each of $\tilde{w}$, $\tilde{w}$, and EC_R , the elasticity is the same across c and μ in the overloaded regime.*

¹⁵Though endogenous congestion from re-dials is undoubtedly bad for the system, the service agents, and the callers themselves, endogenous congestion from re-certifications might actually be considered at least somewhat good: it is a sign that people are successfully receiving SNAP benefits.



(a) Staffing-AHT curves, wait times; decreases in AHT (b) Staffing-AHT curves, procedural denials (c) Wait time vs. recertification length, δ_B (d) Daily procedural denials vs recertification period (changes in δ_B).

Fig. 2. Performance metrics vs staffing levels for different potential changes in call center system design.

Thus, as a main takeaway from the general analysis, these corollaries together show that changes to c and μ have the same respective impact for each of the six performance metrics, suggesting that they have equivalent operational impacts despite their likely quite different costs.

Possible improvements: staffing vs. system design efficiency curves based on Holmes v. Knodell data. While the prior analysis highlights the impact of potential system design changes one at a time, in ?? we leverage the model fitted to *Holmes v. Knodell* call center data to explore the joint efficiency of *simultaneous* improvements in staffing and other system design changes. ?? illustrates in *AHT vs. staffing curves* how performance improvements in wait times and procedural denials depend on interactions between system design and staffing (on the x-axis) improvements. We generate these figures for one of the call centers (#2); results for other call centers are qualitatively similar, hence omitted. We use a θ parameter vector $(\theta_A, \theta_S, \theta_L) = (4, 3, 3)$ — although different values of $\theta_A, \theta_S, \theta_L$ result in different absolute magnitude estimates for wait times, all of our commentary depends on *relative* orderings that remain the same, and procedural denial metrics depend only on θ_A . ??? compare average wait-time ?? and procedural denials on the y-axis as staffing changes, for different improvements in *average handling time (AHT)*; each value of the AHT generates a different staffing curve.¹⁶ In ?? we see wait times decrease sharply at low staffing levels, with diminishing returns at higher staffing. Meanwhile, procedural denials (under our model — total abandonment) decrease linearly in staffing but decreasing AHT increases the rate of enrolled individuals, and hence further improves the *efficiency* of staffing improvements (slope of decrease in denials). Jointly considering system design improvements and staffing increases can *increase the performance improvements* under practically limited additional staffing resources. Conversely, *without* making changes in call center design and service process, it can take an unrealistically large number of additional staff to reduce wait times or procedural denials to acceptable levels.

In ??? we investigate the same types of staffing curves if we consider *longer re-certification cycles*, from 6 months to 1 or 2 years. While longer re-certification cycles don't improve wait-time efficiency as much as improving service time does, longer recertification periods continue to improve procedural denial efficiency. The moderated joint impact on *wait times* reflects an interesting distinction between *per-individual* performance metrics like average wait time per caller vs. *absolute magnitude* performance metrics like total abandonment/procedural denials. Longer re-certification periods maintain enrollment loads but *reduce* arrival rates, therefore averaging waits over a *smaller* total load and

¹⁶System design changes that reduce AHT might include recent interest in AI chatbots, automated responses, increased training for staffing, or self-service options — these could handle simple questions more briefly or redirect callers to new self-service channels.

attenuating the improvements from higher throughput. Interventions that reduce congestion and arrivals have similar *self-attenuating* effects on rate-based rather than absolute metrics.

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Endmatter - Generative AI Usage Statement . We leveraged generative AI (Sonnet 4.5) to help in modifying code, building the Streamlit dashboard web app from simulation code, generating LaTeX tables.

Ethical Considerations Statement. The goal of our work was to improve social impact and equity by carefully studying technical systems (queueing) that currently mediate access to crucial benefits. Our work could have unintended impacts — after all, we focus on introducing and analyzing performance metrics, but there’s always Goodhart’s law, “When a measure becomes a target, it ceases to be a good measure”. Indeed, we refer to prior instances of potential gaming or misreporting/misrepresentation of queueing performance metrics, like with efforts to improve the SSA waits [?]. We try to mitigate potential unintended consequences by highlighting some distinctions between different performance metrics and focusing on a general performance evaluation framework, rather than any single metric in particular.

A Background

Additional discussion on related work. For the most part, the so-called *civic tech* community is the technical practitioner community focused on digital delivery of services and provision of government benefits. Books and grey literature document the challenges of technical deployment and product design in government settings [?]. Recent works in the FAccT community have also explored the interface of benefits technology and equity. [?] audits benefits calculators, which are a category of online tools that distill the government's original, remarkably complex eligibility forms into more UX-friendly, shorter screeners. Such benefits screeners, however, might provide accessibility and reduce frictions at the expense of *accuracy* in eligibility determinations. [?] conducts an interview study to assess SNAP applicants' interactions with potential LLM-based chatbots, studying the interface between trust issues in AI and potential improvements in administrative burdens. [?] find that an efficient advertising budget for GetCalfresh, which streamlines enrollment in California's CalFresh (SNAP), resulted in low takeup among Hispanic individuals: The authors conduct a survey and find general public support for *equitable* advertising budget allocations with higher spending on advertising aimed at harder-to-reach populations.

Additional background on benefits tech. Government digital infrastructure faces several structural challenges that prevent rapid iteration and performance evaluation feedback loops: contracting out implementation to large vendors in cumbersome public procurement processes, larger "waterfall" and non-"agile" contracts that introduce inflexible system design, limited in-house technical capacity, and vendor opacity due to trade secrets protection. For example, when Missouri announced intentions to deploy a new version of Medicaid home-based care determination algorithms, policy analysts and analysts had to band together to bring potentially affected individuals together to audit potential impacts on care. Benefits technological infrastructure (hence called "benefits tech" for short) falls in this category - it must interface with proprietary data mainframes and is often contracted out via public procurement, where few firms have the capacity to work with government legacy systems, and so there is little market pressure to compete on performance. They found that the updated algorithm would remove 66% of affected individuals from crucial home nurse care as well as potential syntax errors in the algorithm [?]. Without this community audit, such deployments would have introduced harmful erroneous determinations silently — it would be three more years before Missouri contracted a consultant to analyze the impacts of an updated algorithm.

The call center system design and deployment in this setting is somewhat different from prior major benefits tech infrastructure projects, in that call center technology is in principle well-developed in the commercial sector. But, call center design and operations are usually contracted out, as Missouri DSS did to Genesys, a major provider. Designing call center systems and queues is a specialized skill with little guidance from FNS as to performance standards and unlikely in-house expertise in social services.

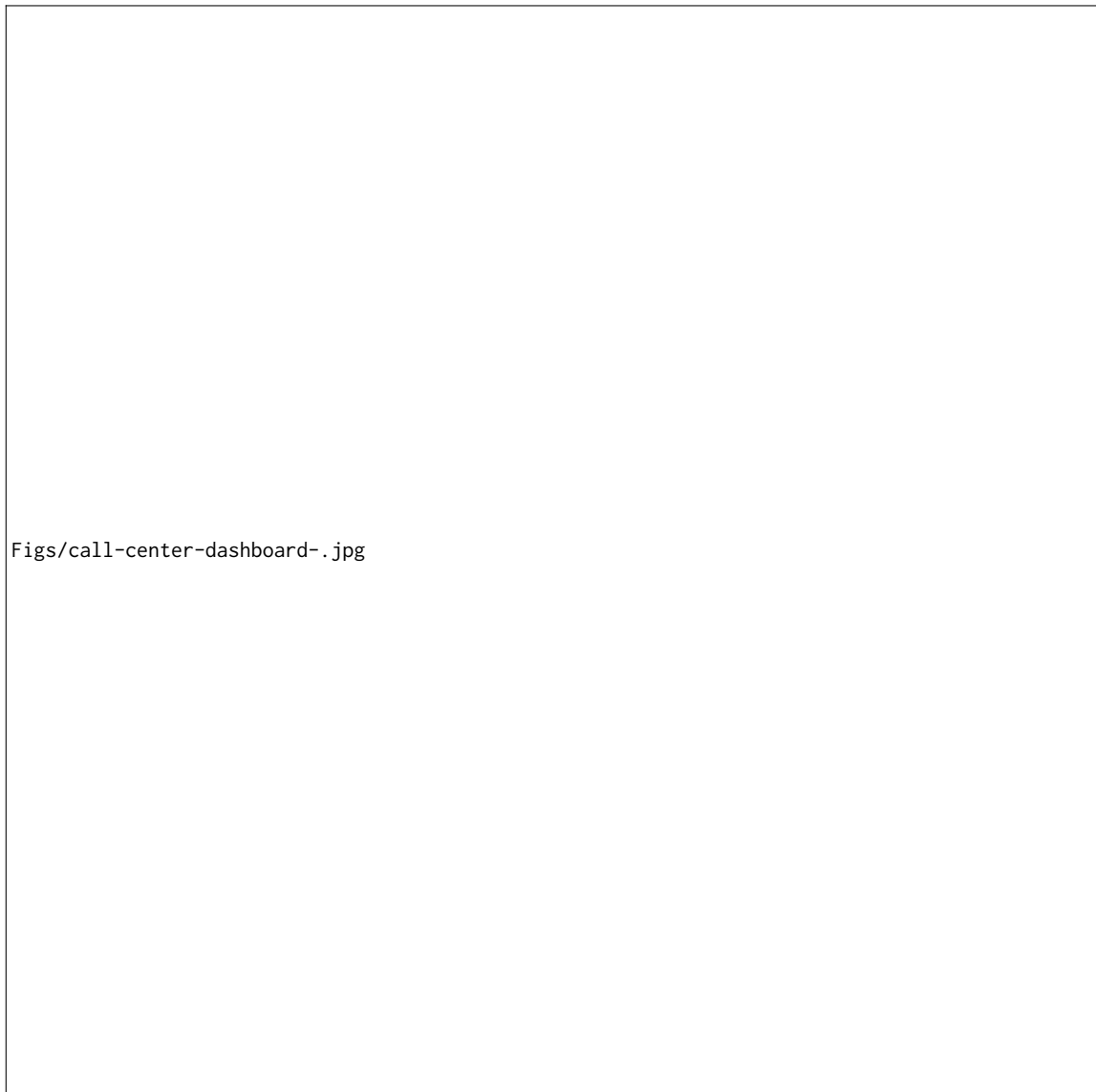


Fig. 3. Screenshot of call center dashboard

B Dashboard

C Call center arrival data

Data Dictionary: Call Center Performance Metrics (Offered, Answered, Abandoned)

Table 2. Key Metrics from Offered, Answered, Abandoned Data Files

Field Name	Description
Offered	Total number of incoming calls presented to the call system or queue.
Answered	Total number of calls successfully answered by an agent.
Abandon	Total number of calls that hung up while waiting in the queue before being answered.
Abandon %	Percentage of calls that abandoned (Abandon / Offered).
Avg Wait / Average Wait	Average time a caller spent waiting in the queue before being connected to an agent or abandoning.
ASA (Average Speed of Answer)	The average time a call is expected to wait in the queue before being answered by an agent.
AHT (Average Handle Time)	The average amount of time an agent spends on an answered call (including talk time and wrap-up work).
Average Abandon Time	The average amount of time abandoned calls spent in the queue before the caller hung up.
Staffing	The total number of staff members scheduled or logged in to handle calls during the interval.
Calls/Person	A staff productivity metric, calculated as calls handled per staff member.

C.1 Parameter fitting based on data

Parameters directly observed in data. From the data, we directly observe some of the parameters.

Recall that we have defined $\hat{\lambda}$ as the total arrival rate of calls, which we observe directly from data. This includes not only the fresh arrivals λ , i.e. calls originating not from prior cases, but also incoming calls from recertifications ($\delta_B \bar{b}$), re-dials from prior abandoned or re-connected calls entering the short orbit ($\delta_S \bar{r}_S$), and re-dials from prior calls entering the long orbit, ($\delta_L \bar{r}_L$).

From data, we also directly observe staffing levels, c , for each call center – we take averages. From SNAP eligibility guidelines, we fix the recertification period is at 6 months (calendar days) ≈ 128.5 model (call center) days (since it's only open weekdays). This fixes the parameter $\delta_B = 1/128.5$.

The wait time data includes Average Handling Time (AHT), and therefore the completion rate of calls, $1/AHT$. Our model further distinguishes service completions into those that complete enrollment (μ_+) vs. those that require a re-connection/re-dial (μ_-). Therefore, completion rates $1/AHT$ are only informative of the sum of these enrollment/re-dial flows:

$$\mu_+ + \mu_- = \frac{MMin}{AHT}$$

To reduce parameter dimensionality, we fix a proportion $\rho \in [0, 1]$ of completed calls that result in completed re-enrollments, hence $\mu_+ = \rho \left(\frac{MMin}{AHT} \right)$. Although ρ is not observed in this data, it may be in other settings, or otherwise informed by the domain.

Parametrizing δ_L from the call center arrival data. The re-dial rate of the long orbit is δ_L , which governs the rate at which re-dials occur from the long orbit (longer timeframe of calling back, for example to obtain and confirm additional

verifications). We fit vector-autoregressive regression (VAR) time-series models to the call center offered and abandoned time series data (adjusting for day-of-week fixed effects). In summary, the empirical time-series analysis surfaces dependence in call center volume (offered and abandoned calls) that operates both over the short-term (dependence on prior 1-2 days) and longer-range dependencies (7-9 days). We therefore set $\delta_L = 1/9$. This is consistent with domain discussion about SNAP interviews and how they often require follow-up due to interview questions that surface or clarify additional documentation needs.

C.2 Call center arrival rate analysis

Table 3. Call Center Statistics

Call Center	Mean Offered	Std	Mean Answered	Std	Mean Abandoned	Std
2	3120.06	2573.98	1229.37	1026.57	1788.04	1872.56
3	2555.30	891.27	882.08	281.89	1500.85	1020.05
4	717.15	211.77	699.21	210.99	17.88	22.54
5	6829.01	1943.26	2729.42	866.85	3880.46	2054.77

Periodicity. We start with one call center, the second in pages 37-87 of exhibit 87. As expected, there are day of week effects in calls offered and abandoned. Call volume is the highest on Monday, and decreases over the week (for both offered and abandoned calls). We adjust for day of week effects by fitting outcomes to a linear model of just day-of-week effects, subtract off the predictions based on just day-of-week and add back the grand mean.

Next we fit a vector autoregressive model to the call center time-series data of offered and abandoned calls, separately for each call center. For ease of interpretation, we don't difference the data, although a Dickey-Fuller test finds only the abandoned calls time series is stationary.

We use 10 lags in the time series analysis. Note that abandoned calls is a proportion of offered calls of the day, and so naturally tracks offered calls.

To summarize the time series dependence patterns: Both the volume of offered and abandoned calls track 1) recent, past 1-2 days of offered and abandoned calls and 2) mid-longer range dependencies, like 7-10 days prior. The number of offered calls has a significant coefficient on lagged offered calls for at least 7 out of the first 9 lags, i.e. persistence of call volume (but not necessarily abandoned calls).

The dependence on prior abandonment is intermittent. Overall, abandonment depends more on prior abandonment than the number of offered calls does. But, we also find occasional statistically significant coefficients on long-range (7-9 day) abandonment.

Table 4. Call center 2

	<i>Dependent variable:</i>	
	(Offered)	(Abandoned)
Adjusted Offered (Lag 1)	0.204*** (0.043)	-0.338*** (0.036)
Adjusted Abandon (Lag 1)	0.603*** (0.053)	1.109*** (0.043)
Adjusted Offered (Lag 2)	0.541*** (0.050)	0.154*** (0.041)
Adjusted Abandon (Lag 2)	-0.466*** (0.065)	-0.090* (0.054)
Adjusted Offered (Lag 3)	0.091* (0.053)	0.123*** (0.043)
Adjusted Abandon (Lag 3)	-0.081 (0.067)	-0.114** (0.055)
Adjusted Offered (Lag 4)	0.114** (0.053)	0.069 (0.043)
Adjusted Abandon (Lag 4)	-0.038 (0.067)	0.035 (0.055)
Adjusted Offered (Lag 5)	0.125** (0.052)	-0.024 (0.043)
Adjusted Abandon (Lag 5)	-0.031 (0.067)	0.125** (0.055)
Adjusted Offered (Lag 6)	-0.121** (0.053)	-0.079* (0.043)
Adjusted Abandon (Lag 6)	-0.065 (0.067)	-0.071 (0.055)
Adjusted Offered (Lag 7)	-0.196*** (0.052)	-0.016 (0.043)
Adjusted Abandon (Lag 7)	0.264*** (0.067)	0.061 (0.055)
Adjusted Offered (Lag 8)	0.076 (0.053)	0.060 (0.043)
Adjusted Abandon (Lag 8)	-0.075 (0.067)	-0.053 (0.055)
Adjusted Offered (Lag 9)	0.285*** (0.052)	0.158*** (0.043)
Adjusted Abandon (Lag 9)	-0.312*** (0.066)	-0.186*** (0.054)
Adjusted Offered (Lag 10)	0.012 (0.050)	-0.086** (0.041)
Adjusted Abandon (Lag 10)	0.071 (0.064)	0.150*** (0.053)
Adjusted Offered (Lag 11)	-0.121*** (0.044)	0.018 (0.036)
Adjusted Abandon (Lag 11)	0.096* (0.053)	-0.039 (0.044)
const	31.047 (20.605)	5.695 (16.878)
Observations	2,076	2,076
R ²	0.952	0.940
Adjusted R ²	0.952	0.939
Residual Std. Error (df = 2053)	562.739	460.946
F Statistic (df = 22; 2053)	1,868.297***	1,457.608***

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 5. Call Center 3

		<i>Dependent variable:</i>	
		(Offered)	(Abandoned)
Adjusted Offered (Lag 1)		−0.091 (0.083)	−0.434*** (0.075)
Adjusted Abandon (Lag 1)		0.886*** (0.092)	1.316*** (0.083)
Adjusted Offered (Lag 2)		0.493*** (0.089)	0.181** (0.081)
Adjusted Abandon (Lag 2)		−0.448*** (0.117)	−0.209** (0.106)
Adjusted Offered (Lag 3)		0.081 (0.094)	−0.009 (0.085)
Adjusted Abandon (Lag 3)		−0.167 (0.119)	−0.059 (0.108)
Adjusted Offered (Lag 4)		0.014 (0.094)	0.028 (0.085)
Adjusted Abandon (Lag 4)		0.048 (0.119)	0.014 (0.108)
Adjusted Offered (Lag 5)		0.123 (0.094)	−0.030 (0.085)
Adjusted Abandon (Lag 5)		0.025 (0.119)	0.230** (0.107)
Adjusted Offered (Lag 6)		0.047 (0.094)	0.108 (0.085)
Adjusted Abandon (Lag 6)		−0.161 (0.118)	−0.222** (0.107)
Adjusted Offered (Lag 7)		−0.079 (0.094)	−0.011 (0.085)
Adjusted Abandon (Lag 7)		0.184 (0.119)	0.081 (0.108)
Adjusted Offered (Lag 8)		0.043 (0.093)	0.038 (0.084)
Adjusted Abandon (Lag 8)		−0.158 (0.118)	−0.152 (0.107)
Adjusted Offered (Lag 9)		0.317*** (0.090)	0.230*** (0.081)
Adjusted Abandon (Lag 9)		−0.270** (0.116)	−0.172 (0.104)
Adjusted Offered (Lag 10)		−0.0003 (0.086)	−0.108 (0.078)
Adjusted Abandon (Lag 10)		0.053 (0.098)	0.129 (0.088)
const		151.875* (78.981)	88.717 (71.421)
Observations		480	480
R ²		0.830	0.897
Adjusted R ²		0.823	0.893
Residual Std. Error (df = 459)		365.760	330.749
F Statistic (df = 20; 459)		112.110***	200.132***

*p<0.1; **p<0.05; ***p<0.01

Table 6. Call Center 4

	<i>Dependent variable:</i>	
	Offered	Abandoned
Adjusted Offered (Lag 1)	0.453*** (0.048)	0.012** (0.005)
Adjusted Abandon (Lag 1)	0.018 (0.478)	0.373*** (0.050)
Adjusted Offered (Lag 2)	0.193*** (0.052)	-0.024*** (0.005)
Adjusted Abandon (Lag 2)	-0.421 (0.502)	0.044 (0.052)
Adjusted Offered (Lag 3)	0.071 (0.055)	0.001 (0.006)
Adjusted Abandon (Lag 3)	0.069 (0.508)	0.610*** (0.053)
Adjusted Offered (Lag 4)	0.065 (0.054)	0.012** (0.006)
Adjusted Abandon (Lag 4)	0.436 (0.924)	-0.159* (0.096)
Adjusted Offered (Lag 5)	-0.017 (0.055)	-0.006 (0.006)
Adjusted Abandon (Lag 5)	0.357 (0.954)	0.066 (0.099)
Adjusted Offered (Lag 6)	-0.089 (0.055)	-0.001 (0.006)
Adjusted Abandon (Lag 6)	-0.788 (0.963)	0.108 (0.100)
Adjusted Offered (Lag 7)	0.006 (0.055)	-0.010* (0.006)
Adjusted Abandon (Lag 7)	0.799 (0.971)	-0.062 (0.101)
Adjusted Offered (Lag 8)	0.065 (0.055)	0.018*** (0.006)
Adjusted Abandon (Lag 8)	0.070 (0.974)	-0.218** (0.101)
Adjusted Offered (Lag 9)	0.113** (0.056)	0.018*** (0.006)
Adjusted Abandon (Lag 9)	-1.014 (0.974)	-0.240** (0.101)
Adjusted Offered (Lag 10)	0.021 (0.056)	-0.009 (0.006)
Adjusted Abandon (Lag 10)	0.403 (0.972)	-0.044 (0.101)
Adjusted Offered (Lag 11)	-0.063 (0.056)	-0.005 (0.006)
Adjusted Abandon (Lag 11)	-0.815 (0.954)	-0.080 (0.099)
Adjusted Offered (Lag 12)	-0.053 (0.056)	0.005 (0.006)
Adjusted Abandon (Lag 12)	0.655 (0.946)	0.214** (0.098)
Adjusted Offered (Lag 13)	0.016 (0.054)	-0.005 (0.006)
Adjusted Abandon (Lag 13)	-1.006 (0.738)	0.321*** (0.076)
Adjusted Offered (Lag 14)	0.085* (0.049)	-0.006 (0.005)
Adjusted Abandon (Lag 14)	0.949 (0.732)	0.024 (0.076)
const	105.124*** (34.398)	1.817 (3.561)
Observations	476	476
R ²	0.531	0.570
Adjusted R ²	0.502	0.543
Residual Std. Error (df = 447)	143.892	14.896
F Statistic (df = 28; 447)	18.106***	21.184***

*p<0.1; **p<0.05; ***p<0.01

Table 7. Call Center 5

		<i>Dependent variable:</i>	
		(Offered)	(Abandoned)
Adjusted Offered (Lag 1)		0.017 (0.099)	-0.402*** (0.098)
Adjusted Abandon (Lag 1)		0.754*** (0.102)	1.183*** (0.100)
Adjusted Offered (Lag 2)		0.586*** (0.107)	0.216** (0.105)
Adjusted Abandon (Lag 2)		-0.489*** (0.119)	-0.126 (0.117)
Adjusted Offered (Lag 3)		0.151 (0.113)	0.142 (0.111)
Adjusted Abandon (Lag 3)		-0.196 (0.124)	-0.212* (0.122)
Adjusted Offered (Lag 4)		0.027 (0.113)	-0.050 (0.111)
Adjusted Abandon (Lag 4)		0.044 (0.123)	0.140 (0.122)
Adjusted Offered (Lag 5)		-0.219* (0.113)	-0.240** (0.111)
Adjusted Abandon (Lag 5)		0.269** (0.123)	0.341*** (0.121)
Adjusted Offered (Lag 6)		0.064 (0.111)	0.038 (0.109)
Adjusted Abandon (Lag 6)		-0.152 (0.122)	-0.129 (0.120)
Adjusted Offered (Lag 7)		-0.021 (0.108)	0.108 (0.107)
Adjusted Abandon (Lag 7)		0.114 (0.118)	-0.060 (0.116)
Adjusted Offered (Lag 8)		0.004 (0.100)	0.002 (0.099)
Adjusted Abandon (Lag 8)		-0.046 (0.111)	-0.001 (0.110)
Adjusted Offered (Lag 9)		0.355*** (0.092)	0.225** (0.090)
Adjusted Abandon (Lag 9)		-0.331*** (0.096)	-0.224** (0.094)
const		343.921** (168.865)	78.282 (166.516)
Observations		481	481
R ²		0.876	0.896
Adjusted R ²		0.871	0.892
Residual Std. Error (df = 462)		673.272	663.906
F Statistic (df = 18; 462)		180.691***	220.763***

*p<0.1; **p<0.05; ***p<0.01

D Analysis

LEMMA D.1. *Suppose the system is in the overloaded regime. Then, the total arrival rate and the total arrival rate without abandonments are given by*

$$\hat{\lambda} = c\mu + \frac{\theta}{\theta_A} \left(\lambda - \frac{\gamma}{\gamma + \delta_B} c\mu_+ \right) \quad \text{and} \quad \tilde{\lambda} = c\mu, \quad (13)$$

respectively.

PROOF. These arrival rate identities follow immediately from the expressions for the steady-state fluid values in the overloaded regime from Corollary ?? and the definitions that $\hat{\lambda} = \lambda + \delta_B \bar{b} + \delta_S \bar{r}_S + \delta_L \bar{r}_L$ and $\tilde{\lambda} = \hat{\lambda} - \theta(\bar{q} - c)$. $\square$

LEMMA D.2. *Suppose the system is in the overloaded regime. Phrased in terms of the procedural denial rate (PD), the mean number of waiting callers ($\bar{q} - c$), mean waiting time ($\bar{w}$), average speed to answer ($\tilde{w}$), the endogenous congestion from re-dials (EC_R), and the endogenous congestion from re-certification (EC_B) can be expressed*

$$\bar{q} - c = \frac{1}{\theta_A} PD, \quad (14)$$

$$\bar{w} = \frac{PD}{c\mu\theta_A + \theta PD}, \quad (15)$$

$$\tilde{w} = \frac{1}{c\mu\theta_A} PD, \quad (16)$$

$$EC_R = c\mu_- + \frac{\theta_S + \theta_L}{\theta_A} PD, \quad (17)$$

$$EC_B = c\mu_+ - \lambda + PD. \quad (18)$$

PROOF. First, let us recall that, in the overloaded regime, the procedural denial rate is given by

$$PD = \lambda - \frac{\gamma}{\gamma + \delta_B} c\mu_+. \quad (19)$$

Hence, by immediate consequence of Lemma ??, the total arrival rate can be phrased in terms of the procedural denials as

$$\hat{\lambda} = c\mu + \frac{\theta}{\theta_A} PD. \quad (20)$$

The proofs of the expressions for the five performance metrics now quickly follow. First, for the mean number of waiting callers, we can observe

$$\bar{q} - c = \frac{1}{\theta_A} \left(\lambda - \frac{\gamma}{\gamma + \delta_B} c\mu_+ \right) = \frac{1}{\theta_A} PD. \quad (21)$$

Accordingly, the waiting time and speed to answer can be written

$$\bar{w} = \frac{\bar{q} - c}{\hat{\lambda}} = \frac{\frac{1}{\theta_A} PD}{c\mu + \frac{\theta}{\theta_A} PD}, \quad (22)$$

and

$$\tilde{w} = \frac{\bar{q} - c}{\tilde{\lambda}} = \frac{\frac{1}{\theta_A} PD}{c\mu}, \quad (23)$$

which both immediately simplify to the stated expressions. Finally, for the endogenous congestion terms, we can draw upon the equilibrium solutions for the overloaded regime in Corollary ?? and find

$$EC_R = \delta_S \bar{r}_S + \delta_L \bar{r}_L = c\mu_- + \frac{\theta_S + \theta_L}{\theta_A} \left(\lambda - \frac{\gamma}{\gamma + \delta_B} c\mu_+ \right) = c\mu_- + \frac{\theta_S + \theta_L}{\theta_A} PD, \quad (24)$$

and

$$EC_B = \delta_B \bar{b} = \frac{\delta_B}{\gamma + \delta_B} c\mu_+ = c\mu_+ - \frac{\gamma}{\gamma + \delta_B} c\mu_+ = c\mu_+ + PD - \lambda, \quad (25)$$

which completes the proof. $\square$

PROPOSITION D.3. *Suppose the system is in the overloaded regime. For the parameters of operational design and control, δ_B , c , μ , and p_+ where $\mu_+ = \mu p_+$ and $\mu_- = \mu(1 - p_+)$, the partial derivatives of the procedural denial rate are*

$$\frac{\partial PD}{\partial \delta_B} = \frac{\gamma c \mu p_+}{(\gamma + \delta_B)^2}, \quad (26)$$

$$\frac{\partial PD}{\partial c} = -\frac{\gamma \mu p_+}{\gamma + \delta_B}, \quad (27)$$

$$\frac{\partial PD}{\partial \mu} = -\frac{\gamma c p_+}{\gamma + \delta_B}, \quad (28)$$

$$\frac{\partial PD}{\partial p_+} = -\frac{\gamma c \mu}{\gamma + \delta_B}. \quad (29)$$

PROOF. Each of these derivatives immediately follows from the expression of the procedural denial rate in terms of the four operational parameters, namely $PD = \lambda - \gamma c \mu p_+ / (\gamma + \delta_B)$. $\square$

PROPOSITION D.4. *Suppose the system is in the overloaded regime. For the parameters of operational design and control, δ_B , c , μ , and p_+ where $\mu_+ = \mu p_+$ and $\mu_- = \mu(1 - p_+)$, the partial derivatives of the mean number of waiting callers can be expressed*

$$\frac{\partial(\bar{q} - c)}{\partial \delta_B} = \frac{\gamma c \mu p_+}{\theta_A (\gamma + \delta_B)^2}, \quad (30)$$

$$\frac{\partial(\bar{q} - c)}{\partial c} = -\frac{\gamma \mu p_+}{\theta_A (\gamma + \delta_B)}, \quad (31)$$

$$\frac{\partial(\bar{q} - c)}{\partial \mu} = -\frac{\gamma c p_+}{\theta_A (\gamma + \delta_B)}, \quad (32)$$

$$\frac{\partial(\bar{q} - c)}{\partial p_+} = -\frac{\gamma c \mu}{\theta_A (\gamma + \delta_B)} \quad (33)$$

PROOF. These expressions follow immediately from Lemma ?? and Proposition ??, as $(\bar{q} - c)$ is linear in PD. $\square$

PROPOSITION D.5. Suppose the system is in the overloaded regime. For the parameters of operational design and control, δ_B , c , μ , and p_+ where $\mu_+ = \mu p_+$ and $\mu_- = \mu(1 - p_+)$, the partial derivatives of the mean waiting time can be expressed

$$\frac{\partial \bar{w}}{\partial \delta_B} = \gamma \theta_A p_+ \left(\frac{c\mu}{(c\mu\theta_A + \theta PD)(\gamma + \delta_B)} \right)^2, \quad (34)$$

$$\frac{\partial \bar{w}}{\partial c} = -\frac{\lambda \mu \theta_A}{(c\mu\theta_A + \theta PD)^2}, \quad (35)$$

$$\frac{\partial \bar{w}}{\partial \mu} = -\frac{\lambda c \theta_A}{(c\mu\theta_A + \theta PD)^2}, \quad (36)$$

$$\frac{\partial \bar{w}}{\partial p_+} = -\frac{\gamma \theta_A}{\gamma + \delta_B} \left(\frac{c\mu}{c\mu\theta_A + \theta PD} \right)^2 \quad (37)$$

PROOF. By Lemma ??, for each $x \in \{\delta_B, c, \mu, p_+\}$, the partial derivative of the mean waiting time can be related to the partial derivative of the procedural denial rate via

$$\frac{\partial \bar{w}}{\partial x} = \frac{\partial}{\partial x} \left(\frac{PD}{c\mu\theta_A + \theta PD} \right) \quad (38)$$

$$= \frac{1}{c\mu\theta_A + \theta PD} \frac{\partial PD}{\partial x} - \frac{PD}{(c\mu\theta_A + \theta PD)^2} \left(\theta_A \frac{\partial(c\mu)}{\partial x} + \theta \frac{\partial PD}{\partial x} \right) \quad (39)$$

$$= \frac{c\mu\theta_A}{(c\mu\theta_A + \theta PD)^2} \frac{\partial PD}{\partial x} - \frac{\theta_A PD}{(c\mu\theta_A + \theta PD)^2} \frac{\partial(c\mu)}{\partial x} \quad (40)$$

$$= \frac{\theta_A}{(c\mu\theta_A + \theta PD)^2} \left(c\mu \frac{\partial PD}{\partial x} - PD \frac{\partial(c\mu)}{\partial x} \right). \quad (41)$$

Hence, by consequence of the partial derivations of PD provided in Proposition ??, we immediately obtain the partial derivatives of $\bar{w}$. $\square$

PROPOSITION D.6. Suppose the system is in the overloaded regime. For the parameters of operational design and control, δ_B , c , μ , and p_+ where $\mu_+ = \mu p_+$ and $\mu_- = \mu(1 - p_+)$, the partial derivatives of the average speed to answer can be expressed

$$\frac{\partial \tilde{w}}{\partial \delta_B} = \frac{\gamma p_+}{\theta_A(\gamma + \delta_B)^2}, \quad (42)$$

$$\frac{\partial \tilde{w}}{\partial c} = -\frac{\gamma p_+}{c\theta_A(\gamma + \delta_B)} - \frac{PD}{c^2\mu\theta_A}, \quad (43)$$

$$\frac{\partial \tilde{w}}{\partial \mu} = -\frac{\gamma p_+}{\mu\theta_A(\gamma + \delta_B)} - \frac{PD}{c\mu^2\theta_A}, \quad (44)$$

$$\frac{\partial \tilde{w}}{\partial p_+} = -\frac{\gamma}{\theta_A(\gamma + \delta_B)}. \quad (45)$$

PROOF. From Lemma ??, we have that for $x \in \{c, \mu\}$,

$$\frac{\partial \tilde{w}}{\partial x} = \frac{1}{c\mu\theta_A} \frac{\partial PD}{\partial x} + \frac{PD}{\theta_A} \frac{\partial}{\partial x} \left(\frac{1}{c\mu} \right), \quad (46)$$

and, for $x \in \{\delta_B, p_+\}$, the partial derivative of $\tilde{w}$ is simply that of PD divided by $c\mu\theta_A$. Hence, via Proposition ??, we achieve the stated expressions. $\square$

PROPOSITION D.7. Suppose the system is in the overloaded regime. For the parameters of operational design and control, δ_B , c , μ , and p_+ where $\mu_+ = \mu p_+$ and $\mu_- = \mu(1 - p_+)$, the partial derivatives of the endogenous congestion from re-dials can

be expressed

$$\frac{\partial \text{EC}_R}{\partial \delta_B} = \frac{\gamma c \mu p_+ (\theta_S + \theta_L)}{\theta_A (\gamma + \delta_B)^2}, \quad (47)$$

$$\frac{\partial \text{EC}_R}{\partial c} = \mu(1 - p_+) - \frac{\gamma \mu p_+ (\theta_S + \theta_L)}{\theta_A (\gamma + \delta_B)}, \quad (48)$$

$$\frac{\partial \text{EC}_R}{\partial \mu} = c(1 - p_+) - \frac{\gamma c p_+ (\theta_S + \theta_L)}{\theta_A (\gamma + \delta_B)}, \quad (49)$$

$$\frac{\partial \text{EC}_R}{\partial p_+} = -c\mu - \frac{\gamma c \mu (\theta_S + \theta_L)}{\theta_A (\gamma + \delta_B)} \quad (50)$$

PROOF. By Lemma ??, the partial derivative of the re-dial endogenous congestion is

$$\frac{\partial \text{EC}_R}{\partial x} = \frac{\partial}{\partial x} (c\mu(1 - p_+)) + \frac{\theta_S + \theta_L}{\theta_A} \frac{\partial \text{PD}}{\partial x} \quad (51)$$

for each $x \in \{\delta_B, c, \mu, p_+\}$. Hence, by Proposition ??, we immediately achieve the stated results. $\square$

PROPOSITION D.8. Suppose the system is in the overloaded regime. For the parameters of operational design and control, δ_B , c , μ , and p_+ where $\mu_+ = \mu p_+$ and $\mu_- = \mu(1 - p_+)$, the partial derivatives of the endogenous congestion from re-certifications can be expressed

$$\frac{\partial \text{EC}_B}{\partial \delta_B} = \frac{\gamma c \mu p_+}{(\gamma + \delta_B)^2}, \quad (52)$$

$$\frac{\partial \text{EC}_B}{\partial c} = \frac{\delta_B \mu p_+}{\gamma + \delta_B}, \quad (53)$$

$$\frac{\partial \text{EC}_B}{\partial \mu} = \frac{\delta_B c p_+}{\gamma + \delta_B}, \quad (54)$$

$$\frac{\partial \text{EC}_B}{\partial p_+} = \frac{\delta_B c \mu}{\gamma + \delta_B}. \quad (55)$$

PROOF. Lemma ?? implies that the partial derivative of the re-certification endogenous congestion is simply a sum of two partial derivatives,

$$\frac{\partial \text{EC}_B}{\partial x} = \frac{\partial}{\partial x} (c\mu p_+) + \frac{\partial \text{PD}}{\partial x}, \quad (56)$$

and from this relationship, we immediately achieve the proof via simplifying from Proposition ??. $\square$

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